

Nirma University
Institute of Technology
Semester End Examination (IR), January - 2023
Master of Computer Applications, Semester - I
6CS103 Operating System

Roll/Exam No

Supervisor's Initial

Time: 3 Hrs

Max. Marks: 100

Instructions:

1. Attempt all questions. (No Optional Questions)
2. Figures to right indicate full marks.
3. Draw neat sketches wherever necessary.
4. Make suitable assumptions wherever necessary and specify them.
5. Section-wise separate answer book should be used

SECTION - I

Q.1 Answer the following questions. [20]

- A.** Discuss briefly the four main elements of a typical computer system. [7]
CO1, BL1 Draw neat diagram showing how these elements are connected with each other.
- B.** What is a 'kernel' in context of operating system? Differentiate [7]
CO1, BL2 monolithic kernel Vs microkernel. Also give one example of OS in which monolithic and microkernel is present.
- C.** What is a multiprocessor system? Discuss SMP (Symmetric Multi- [6]
CO1, BL1 processor) in detail with suitable diagram.

Q.2 Answer the following questions. [20]

- A.** What is meant by 'Process Image'? Discuss briefly various elements [7]
CO2, BL1 of a process image.
- B.** What is meant by 'Thread' in context of operating system? Illustrate [7]
CO2, BL4 and describe the single threaded and multi-threaded process model with the help of an appropriate diagram.
- C.** Compare and contrast User level threads (ULT) with Kernel level [6]
CO2, BL3 threads (KLT).

Q.3 List three degrees of awareness between processes in context of [10]
CO3, BL3 process interaction and briefly define each with respect to relationship among processes, influence that one process has on other and potential problem that may exist in each of the case.

SECTION-II

- Q.4 Answer the following questions. [20]**
- A.** What is deadlock? Discuss the three necessary conditions that must be present for deadlock to be possible? Also, mention and discuss the fourth essential condition that create deadlock. [7]
CO2, BL4
- B.** Given the following memory partitions. [7]
CO2, BL6 9M, 13M, 23M, 19M, 9M, 7M, 15M, 37M (in order).
Initially the partitions are empty. Show the placement of processes with size 7M, 19M, 9M (in order) for **first fit** and **best fit**.
Out of first fit and best fit, which one is better with respect to the above example?
- C.** State the classic producer/consumer problem. Device a pseudo-code for solving infinite-buffer producer/consumer problem using binary semaphore[s]. [6]
CO2, BL5
- Q.5 Answer the following questions. [20]**
- A.** Discuss the following in context of memory management taking suitable example: [7]
CO4, BL3 (i) Logical address
(ii) Relative address
(iii) Physical address
- B.** What is the meant by 'rotational latency' in context of disk management? Show with the help of diagram the order in which the requests will be satisfied for FIFO (First-in-first-out) and SSTF (Shortest service time first) disk scheduling algorithms for the following sequence of disk track requests. [7]
CO4, BL6 118, 62, 8, 39, 145, 184, 108, 127, 25.
Assume the disk head is initially at track no. 18.
- C.** Describe 'Bankers' algorithm of deadlock avoidance with the help of a suitable example. [6]
CO4, BL3
- Q.6 Answer the following questions. [10]**
- A.** What is the difference between a 'preemptive' and non-preemptive scheduling algorithm? [6]
CO3, BL4 Explain Shortest Process Next (SPN) scheduling algorithm with appropriate example.
- B.** Consider a hypothetical 32-bit microprocessor having 32-bit instructions composed of two fields. The first byte contains the opcode and the remainder an immediate operand or an operand address. [4]
CO3, BL5 How many bits are needed for the program counter (PC) and the instruction register (IR)? Give reason for your answer.

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